



МИНИСТЕРСТВО НАУКИ  
И ВЫСШЕГО ОБРАЗОВАНИЯ  
РОССИЙСКОЙ ФЕДЕРАЦИИ

Федеральное государственное бюджетное  
образовательное учреждение высшего образования  
«НОВОСИБИРСКИЙ ГОСУДАРСТВЕННЫЙ ТЕХНИЧЕСКИЙ УНИВЕРСИТЕТ»



**НГТУ  
НЭТИ** | **Факультет прикладной  
математики и информатики**

Кафедра теоретической и прикладной информатики  
Лабораторная работа № 1  
по дисциплине «Методы оптимизации»

## **МЕТОДЫ ОДНОМЕРНОГО ПОИСКА.**

|              |                    |
|--------------|--------------------|
| Бригада 9    | ГРУШЕВ АНДРЕЙ      |
| Группа ПМ-05 | ХАБАРОВА АНАСТАСИЯ |
| Вариант 9    |                    |

|               |                              |
|---------------|------------------------------|
| Преподаватели | ЛЕМЕШКО БОРИС ЮРЬЕВИЧ        |
|               | ФИЛИППОВА ЕЛЕНА ВЛАДИМИРОВНА |

Новосибирск, 2023

## Цель работы

Ознакомиться с методами одномерного поиска, используемыми в многомерных методах минимизации функций  $n$  переменных. Сравнить различные алгоритмы по эффективности на тестовых примерах.

## Задание

1. Реализовать методы дихотомии, золотого сечения, исследовать их сходимость и провести сравнение по числу вычислений функции для достижения заданной точности  $\varepsilon$  от  $10^{-1}$  до  $10^{-7}$ . Построить график зависимости количества вычислений минимизируемой функции от десятичного логарифма задаваемой точности  $\varepsilon$ .
2. Реализовать алгоритм поиска интервала, содержащего минимум функции.
3. Реализовать метод Фибоначчи, сравнить его с методами дихотомии и золотого сечения.

Минимизируемая функция:  $y(x) = (x - 9)^2, x \in [-2, 20]$ .

## Результаты исследований

Метод дихотомии для  $y(x) = (x - 9)^2$ :

| $i$ | $x_1$        | $x_2$        | $f(x_1)$     | $f(x_2)$     | $a_i$         | $b_i$        | $b_i - a_i$  | $\frac{b_{i-1} - a_{i-1}}{b_i - a_i}$ |
|-----|--------------|--------------|--------------|--------------|---------------|--------------|--------------|---------------------------------------|
| 1   | 9.000000E+00 | 9.000000E+00 | 2.499912E-21 | 2.499912E-21 | -2.000000E+00 | 2.000000E+01 | 2.200000E+01 | 0                                     |

Метод дихотомии для  $y(x) = (x - 2)^2$ :

| $i$ | $x_1$        | $x_2$        | $f(x_1)$     | $f(x_2)$     | $a_i$         | $b_i$        | $b_i - a_i$  | $\frac{b_{i-1} - a_{i-1}}{b_i - a_i}$ |
|-----|--------------|--------------|--------------|--------------|---------------|--------------|--------------|---------------------------------------|
| 1   | 9.000000E+00 | 9.000000E+00 | 4.900000E+01 | 4.900000E+01 | -2.000000E+00 | 2.000000E+01 | 2.200000E+01 | 0                                     |
| 2   | 3.500000E+00 | 3.500000E+00 | 2.250000E+00 | 2.250000E+00 | -2.000000E+00 | 9.000000E+00 | 1.100000E+01 | 2.000000                              |
| 3   | 7.500000E-01 | 7.500000E-01 | 1.562500E+00 | 1.562500E+00 | -2.000000E+00 | 3.500000E+00 | 5.500000E+00 | 2.000000                              |
| 4   | 2.125000E+00 | 2.125000E+00 | 1.562500E-02 | 1.562500E-02 | 7.500000E-01  | 3.500000E+00 | 2.750000E+00 | 2.000000                              |
| 5   | 1.437500E+00 | 1.437500E+00 | 3.164063E-01 | 3.164062E-01 | 7.500000E-01  | 2.125000E+00 | 1.375000E+00 | 2.000000                              |
| 6   | 1.781250E+00 | 1.781250E+00 | 4.785156E-02 | 4.785156E-02 | 1.437500E+00  | 2.125000E+00 | 6.875000E-01 | 2.000000                              |
| 7   | 1.953125E+00 | 1.953125E+00 | 2.197266E-03 | 2.197266E-03 | 1.781250E+00  | 2.125000E+00 | 3.437500E-01 | 2.000000                              |
| 8   | 2.039062E+00 | 2.039063E+00 | 1.525879E-03 | 1.525879E-03 | 1.953125E+00  | 2.125000E+00 | 1.718750E-01 | 2.000000                              |
| 9   | 1.996094E+00 | 1.996094E+00 | 1.525879E-05 | 1.525879E-05 | 1.953125E+00  | 2.039063E+00 | 8.593750E-02 | 2.000000                              |
| 10  | 2.017578E+00 | 2.017578E+00 | 3.089905E-04 | 3.089905E-04 | 1.996094E+00  | 2.039063E+00 | 4.296875E-02 | 2.000000                              |
| 11  | 2.006836E+00 | 2.006836E+00 | 4.673004E-05 | 4.673004E-05 | 1.996094E+00  | 2.017578E+00 | 2.148438E-02 | 2.000000                              |
| 12  | 2.001465E+00 | 2.001465E+00 | 2.145767E-06 | 2.145767E-06 | 1.996094E+00  | 2.006836E+00 | 1.074219E-02 | 2.000000                              |
| 13  | 1.998779E+00 | 1.998779E+00 | 1.490116E-06 | 1.490116E-06 | 1.996094E+00  | 2.001465E+00 | 5.371094E-03 | 2.000000                              |
| 14  | 2.000122E+00 | 2.000122E+00 | 1.490116E-08 | 1.490116E-08 | 1.998779E+00  | 2.001465E+00 | 2.685547E-03 | 2.000000                              |
| 15  | 1.999451E+00 | 1.999451E+00 | 3.017485E-07 | 3.017484E-07 | 1.998779E+00  | 2.000122E+00 | 1.342774E-03 | 2.000000                              |
| 16  | 1.999786E+00 | 1.999786E+00 | 4.563481E-08 | 4.563477E-08 | 1.999451E+00  | 2.000122E+00 | 6.713868E-04 | 2.000000                              |
| 17  | 1.999954E+00 | 1.999954E+00 | 2.095477E-09 | 2.095468E-09 | 1.999786E+00  | 2.000122E+00 | 3.356935E-04 | 2.000000                              |
| 18  | 2.000038E+00 | 2.000038E+00 | 1.455190E-09 | 1.455198E-09 | 1.999954E+00  | 2.000122E+00 | 1.678468E-04 | 1.999999                              |
| 19  | 1.999996E+00 | 1.999996E+00 | 1.455205E-11 | 1.455129E-11 | 1.999954E+00  | 2.000038E+00 | 8.392344E-05 | 1.999999                              |
| 20  | 2.000017E+00 | 2.000017E+00 | 2.946757E-10 | 2.946791E-10 | 1.999996E+00  | 2.000038E+00 | 4.196177E-05 | 1.999998                              |
| 21  | 2.000007E+00 | 2.000007E+00 | 4.456500E-11 | 4.456633E-11 | 1.999996E+00  | 2.000017E+00 | 2.098093E-05 | 1.999995                              |
| 22  | 2.000001E+00 | 2.000001E+00 | 2.046311E-12 | 2.046597E-12 | 1.999996E+00  | 2.000007E+00 | 1.049052E-05 | 1.999990                              |
| 23  | 1.999999E+00 | 1.999999E+00 | 1.421129E-12 | 1.420890E-12 | 1.999996E+00  | 2.000001E+00 | 5.245309E-06 | 1.999981                              |
| 24  | 2.000000E+00 | 2.000000E+00 | 1.420652E-14 | 1.423037E-14 | 1.999999E+00  | 2.000001E+00 | 2.622704E-06 | 1.999962                              |
| 25  | 1.999999E+00 | 1.999999E+00 | 2.877893E-13 | 2.876820E-13 | 1.999999E+00  | 2.000000E+00 | 1.311402E-06 | 1.999924                              |
| 26  | 2.000000E+00 | 2.000000E+00 | 4.352833E-14 | 4.348661E-14 | 1.999999E+00  | 2.000000E+00 | 6.557511E-07 | 1.999848                              |
| 27  | 2.000000E+00 | 2.000000E+00 | 2.000027E-15 | 1.991093E-15 | 2.000000E+00  | 2.000000E+00 | 3.279255E-07 | 1.999695                              |
| 28  | 2.000000E+00 | 2.000000E+00 | 1.386424E-15 | 1.393881E-15 | 2.000000E+00  | 2.000000E+00 | 1.640128E-07 | 1.999390                              |

Метод золотого сечения для  $y(x) = (x - 9)^2$ :

| $i$ | $x_1$        | $x_2$        | $f(x_1)$     | $f(x_2)$     | $a_i$         | $b_i$        | $b_i - a_i$  | $\frac{b_{i-1} - a_{i-1}}{b_i - a_i}$ |
|-----|--------------|--------------|--------------|--------------|---------------|--------------|--------------|---------------------------------------|
| 1   | 6.403252E+00 | 1.159675E+01 | 6.743099E+00 | 6.743099E+00 | -2.000000E+00 | 2.000000E+01 | 2.200000E+01 | 0                                     |
| 2   | 1.159675E+01 | 1.480650E+01 | 6.743099E+00 | 3.371549E+01 | 6.403252E+00  | 2.000000E+01 | 1.359675E+01 | 1.618034                              |
| 3   | 9.613009E+00 | 1.159675E+01 | 3.757800E-01 | 6.743099E+00 | 6.403252E+00  | 1.480650E+01 | 8.403252E+00 | 1.618034                              |
| 4   | 8.386991E+00 | 9.613009E+00 | 3.757800E-01 | 3.757800E-01 | 6.403252E+00  | 1.159675E+01 | 5.193496E+00 | 1.618034                              |
| 5   | 9.613009E+00 | 1.037073E+01 | 3.757800E-01 | 1.878900E+00 | 8.386991E+00  | 1.159675E+01 | 3.209757E+00 | 1.618034                              |
| 6   | 9.144712E+00 | 9.613009E+00 | 2.094150E-02 | 3.757800E-01 | 8.386991E+00  | 1.037073E+01 | 1.983739E+00 | 1.618034                              |
| 7   | 8.855288E+00 | 9.144712E+00 | 2.094150E-02 | 2.094150E-02 | 8.386991E+00  | 9.613009E+00 | 1.226018E+00 | 1.618034                              |
| 8   | 9.144712E+00 | 9.323585E+00 | 2.094150E-02 | 1.047075E-01 | 8.855288E+00  | 9.613009E+00 | 7.577208E-01 | 1.618034                              |
| 9   | 9.034162E+00 | 9.144712E+00 | 1.167030E-03 | 2.094150E-02 | 8.855288E+00  | 9.323585E+00 | 4.682972E-01 | 1.618034                              |

|    |              |              |              |              |              |              |              |          |
|----|--------------|--------------|--------------|--------------|--------------|--------------|--------------|----------|
| 10 | 8.965838E+00 | 9.034162E+00 | 1.167030E-03 | 1.167030E-03 | 8.855288E+00 | 9.144712E+00 | 2.894236E-01 | 1.618034 |
| 11 | 9.034162E+00 | 9.076388E+00 | 1.167030E-03 | 5.835150E-03 | 8.965838E+00 | 9.144712E+00 | 1.788736E-01 | 1.618034 |
| 12 | 9.008065E+00 | 9.034162E+00 | 6.503631E-05 | 1.167030E-03 | 8.965838E+00 | 9.076388E+00 | 1.105500E-01 | 1.618034 |
| 13 | 8.991935E+00 | 9.008065E+00 | 6.503640E-05 | 6.503631E-05 | 8.965838E+00 | 9.034162E+00 | 6.832364E-02 | 1.618034 |
| 14 | 9.008065E+00 | 9.018033E+00 | 6.503631E-05 | 3.251816E-04 | 8.991935E+00 | 9.034162E+00 | 4.222633E-02 | 1.618034 |
| 15 | 9.001904E+00 | 9.008065E+00 | 3.624338E-06 | 6.503631E-05 | 8.991935E+00 | 9.018033E+00 | 2.609731E-02 | 1.618034 |
| 16 | 8.998096E+00 | 9.001904E+00 | 3.624363E-06 | 3.624338E-06 | 8.991935E+00 | 9.008065E+00 | 1.612902E-02 | 1.618034 |
| 17 | 9.001904E+00 | 9.004257E+00 | 3.624338E-06 | 1.812174E-05 | 8.998096E+00 | 9.008065E+00 | 9.968285E-03 | 1.618034 |
| 18 | 9.000449E+00 | 9.001904E+00 | 2.019757E-07 | 3.624338E-06 | 8.998096E+00 | 9.004257E+00 | 6.160739E-03 | 1.618034 |
| 19 | 8.999551E+00 | 9.000449E+00 | 2.019810E-07 | 2.019757E-07 | 8.998096E+00 | 9.001904E+00 | 3.807545E-03 | 1.618034 |
| 20 | 9.000449E+00 | 9.001005E+00 | 2.019757E-07 | 1.009884E-06 | 8.999551E+00 | 9.001904E+00 | 2.353192E-03 | 1.618034 |
| 21 | 9.000106E+00 | 9.000449E+00 | 1.125517E-08 | 2.019757E-07 | 8.999551E+00 | 9.001005E+00 | 1.454353E-03 | 1.618034 |
| 22 | 8.999894E+00 | 9.000106E+00 | 1.125650E-08 | 1.125517E-08 | 8.999551E+00 | 9.000449E+00 | 8.988400E-04 | 1.618033 |
| 23 | 9.000106E+00 | 9.000237E+00 | 1.125517E-08 | 5.627793E-08 | 8.999894E+00 | 9.000449E+00 | 5.555137E-04 | 1.618034 |
| 24 | 9.000025E+00 | 9.000106E+00 | 6.271187E-10 | 1.125517E-08 | 8.999894E+00 | 9.000237E+00 | 3.433263E-04 | 1.618034 |
| 25 | 8.999975E+00 | 9.000025E+00 | 6.274232E-10 | 6.271187E-10 | 8.999894E+00 | 9.000106E+00 | 2.121870E-04 | 1.618036 |
| 26 | 9.000025E+00 | 9.000056E+00 | 6.271187E-10 | 3.135980E-09 | 8.999975E+00 | 9.000106E+00 | 1.311388E-04 | 1.618034 |
| 27 | 9.000006E+00 | 9.000025E+00 | 3.491929E-11 | 6.271187E-10 | 8.999975E+00 | 9.000056E+00 | 8.104824E-05 | 1.618034 |
| 28 | 8.999994E+00 | 9.000006E+00 | 3.499257E-11 | 3.491929E-11 | 8.999975E+00 | 9.000025E+00 | 5.009076E-05 | 1.618028 |
| 29 | 9.000006E+00 | 9.000013E+00 | 3.491929E-11 | 1.747027E-10 | 8.999994E+00 | 9.000025E+00 | 3.095779E-05 | 1.618034 |
| 30 | 9.000001E+00 | 9.000006E+00 | 1.939589E-12 | 3.491929E-11 | 8.999994E+00 | 9.000013E+00 | 1.913297E-05 | 1.618034 |
| 31 | 8.999999E+00 | 9.000001E+00 | 1.956685E-12 | 1.939589E-12 | 8.999994E+00 | 9.000006E+00 | 1.182471E-05 | 1.61805  |
| 32 | 9.000003E+00 | 9.000003E+00 | 1.939589E-12 | 9.720804E-12 | 8.999999E+00 | 9.000006E+00 | 7.308070E-06 | 1.618034 |
| 33 | 9.000000E+00 | 9.000001E+00 | 1.065278E-13 | 1.939589E-12 | 8.999999E+00 | 9.000003E+00 | 4.516636E-06 | 1.618034 |
| 34 | 9.000000E+00 | 9.000000E+00 | 1.105927E-13 | 1.065278E-13 | 8.999999E+00 | 9.000001E+00 | 2.791507E-06 | 1.617992 |
| 35 | 9.000000E+00 | 9.000001E+00 | 1.065278E-13 | 5.383245E-13 | 9.000000E+00 | 9.000001E+00 | 1.725246E-06 | 1.618034 |
| 36 | 9.000000E+00 | 9.000000E+00 | 5.583170E-15 | 1.065278E-13 | 9.000000E+00 | 9.000001E+00 | 1.066261E-06 | 1.618034 |
| 37 | 9.000000E+00 | 9.000000E+00 | 6.538634E-15 | 5.583170E-15 | 9.000000E+00 | 9.000000E+00 | 6.589406E-07 | 1.618144 |
| 38 | 9.000000E+00 | 9.000000E+00 | 5.583170E-15 | 2.918327E-14 | 9.000000E+00 | 9.000000E+00 | 4.072477E-07 | 1.618034 |
| 39 | 9.000000E+00 | 9.000000E+00 | 2.333660E-16 | 5.583170E-15 | 9.000000E+00 | 9.000000E+00 | 2.516929E-07 | 1.618034 |
| 40 | 9.000000E+00 | 9.000000E+00 | 4.594430E-16 | 2.333660E-16 | 9.000000E+00 | 9.000000E+00 | 1.555824E-07 | 1.617746 |

Метод золотого сечения для  $y(x) = (x - 2)^2$ :

| $i$ | $x_1$         | $x_2$        | $f(x_1)$     | $f(x_2)$     | $a_i$         | $b_i$        | $b_i - a_i$  | $\frac{b_{i-1} - a_{i-1}}{b_i - a_i}$ |
|-----|---------------|--------------|--------------|--------------|---------------|--------------|--------------|---------------------------------------|
| 1   | 6.403252E+00  | 1.159675E+01 | 1.938863E+01 | 9.209757E+01 | -2.000000E+00 | 2.000000E+01 | 2.200000E+01 | 0                                     |
| 2   | 3.193496E+00  | 6.403252E+00 | 1.424432E+00 | 1.938863E+01 | -2.000000E+00 | 1.159675E+01 | 1.359675E+01 | 1.618034                              |
| 3   | 1.209757E+00  | 3.193496E+00 | 6.244844E-01 | 1.424432E+00 | -2.000000E+00 | 6.403252E+00 | 8.403252E+00 | 1.618034                              |
| 4   | -1.626124E-02 | 1.209757E+00 | 4.065309E+00 | 6.244844E-01 | -2.000000E+00 | 3.193496E+00 | 5.193496E+00 | 1.618034                              |
| 5   | 1.209757E+00  | 1.967478E+00 | 6.244844E-01 | 1.057711E-03 | -1.626124E-02 | 3.193496E+00 | 3.209757E+00 | 1.618034                              |
| 6   | 1.967478E+00  | 2.435775E+00 | 1.057711E-03 | 1.898996E-01 | 1.209757E+00  | 3.193496E+00 | 1.983739E+00 | 1.618034                              |
| 7   | 1.678054E+00  | 1.967478E+00 | 1.036493E-01 | 1.057711E-03 | 1.209757E+00  | 2.435775E+00 | 1.226018E+00 | 1.618034                              |
| 8   | 1.967478E+00  | 2.146351E+00 | 1.057711E-03 | 2.141865E-02 | 1.678054E+00  | 2.435775E+00 | 7.577208E-01 | 1.618034                              |
| 9   | 1.856928E+00  | 1.967478E+00 | 2.046973E-02 | 1.057711E-03 | 1.678054E+00  | 2.146351E+00 | 4.682972E-01 | 1.618034                              |
| 10  | 1.967478E+00  | 2.035801E+00 | 1.057711E-03 | 1.281723E-03 | 1.856928E+00  | 2.146351E+00 | 2.894236E-01 | 1.618034                              |
| 11  | 1.925251E+00  | 1.967478E+00 | 5.587385E-03 | 1.057711E-03 | 1.856928E+00  | 2.035801E+00 | 1.788736E-01 | 1.618034                              |
| 12  | 1.967478E+00  | 1.993575E+00 | 1.057711E-03 | 4.128280E-05 | 1.925251E+00  | 2.035801E+00 | 1.105500E-01 | 1.618034                              |
| 13  | 1.993575E+00  | 2.009704E+00 | 4.128280E-05 | 9.416480E-05 | 1.967478E+00  | 2.035801E+00 | 6.832364E-02 | 1.618034                              |
| 14  | 1.983607E+00  | 1.993575E+00 | 2.687453E-04 | 4.128280E-05 | 1.967478E+00  | 2.009704E+00 | 4.222633E-02 | 1.618034                              |
| 15  | 1.993575E+00  | 1.999736E+00 | 4.128280E-05 | 6.992293E-08 | 1.983607E+00  | 2.009704E+00 | 2.609731E-02 | 1.618034                              |
| 16  | 1.999736E+00  | 2.003543E+00 | 6.992293E-08 | 1.255367E-05 | 1.993575E+00  | 2.009704E+00 | 1.612902E-02 | 1.618034                              |
| 17  | 1.997382E+00  | 1.999736E+00 | 6.851951E-06 | 6.992293E-08 | 1.993575E+00  | 2.003543E+00 | 9.968285E-03 | 1.618034                              |
| 18  | 1.999736E+00  | 2.001190E+00 | 6.992293E-08 | 1.415917E-06 | 1.997382E+00  | 2.003543E+00 | 6.160739E-03 | 1.618034                              |
| 19  | 1.998837E+00  | 1.999736E+00 | 1.353197E-06 | 6.992293E-08 | 1.997382E+00  | 2.001190E+00 | 3.807546E-03 | 1.618034                              |
| 20  | 1.999736E+00  | 2.000291E+00 | 6.992293E-08 | 8.472944E-08 | 1.998837E+00  | 2.001190E+00 | 2.353193E-03 | 1.618034                              |
| 21  | 1.999392E+00  | 1.999736E+00 | 3.693680E-07 | 6.992293E-08 | 1.998837E+00  | 2.000291E+00 | 1.454353E-03 | 1.618034                              |
| 22  | 1.999736E+00  | 1.999948E+00 | 6.992293E-08 | 2.729332E-09 | 1.999392E+00  | 2.000291E+00 | 8.988397E-04 | 1.618034                              |
| 23  | 1.999948E+00  | 2.000079E+00 | 2.729332E-09 | 6.224620E-09 | 1.999736E+00  | 2.000291E+00 | 5.555127E-04 | 1.618036                              |
| 24  | 1.999867E+00  | 1.999948E+00 | 1.776641E-08 | 2.729332E-09 | 1.999736E+00  | 2.000079E+00 | 3.433257E-04 | 1.618034                              |
| 25  | 1.999948E+00  | 1.999998E+00 | 2.729332E-09 | 4.630851E-12 | 1.999867E+00  | 2.000079E+00 | 2.121870E-04 | 1.618034                              |
| 26  | 1.999998E+00  | 2.000029E+00 | 4.630851E-12 | 8.297580E-10 | 1.999948E+00  | 2.000079E+00 | 1.311393E-04 | 1.618028                              |
| 27  | 1.999979E+00  | 1.999998E+00 | 4.530608E-10 | 4.630851E-12 | 1.999948E+00  | 2.000029E+00 | 8.104853E-05 | 1.618034                              |
| 28  | 1.999998E+00  | 2.000010E+00 | 4.630851E-12 | 9.355838E-11 | 1.999979E+00  | 2.000029E+00 | 5.009074E-05 | 1.618034                              |
| 29  | 1.999991E+00  | 1.999998E+00 | 8.949924E-11 | 4.630851E-12 | 1.999979E+00  | 2.000010E+00 | 3.095778E-05 | 1.618034                              |
| 30  | 1.999998E+00  | 2.000002E+00 | 4.630851E-12 | 5.590467E-12 | 1.999991E+00  | 2.000010E+00 | 1.913296E-05 | 1.618034                              |
| 31  | 1.999995E+00  | 1.999998E+00 | 2.444041E-11 | 4.630851E-12 | 1.999991E+00  | 2.000002E+00 | 1.182482E-05 | 1.618034                              |
| 32  | 1.999998E+00  | 2.000000E+00 | 4.630851E-12 | 1.823671E-13 | 1.999995E+00  | 2.000002E+00 | 7.308141E-06 | 1.618034                              |
| 33  | 2.000000E+00  | 2.000001E+00 | 1.823671E-13 | 4.087321E-13 | 1.999998E+00  | 2.000002E+00 | 4.516358E-06 | 1.618149                              |
| 34  | 1.999999E+00  | 2.000000E+00 | 1.178905E-12 | 1.823671E-13 | 1.999998E+00  | 2.000001E+00 | 2.791263E-06 | 1.618034                              |
| 35  | 2.000000E+00  | 2.000000E+00 | 1.823671E-13 | 3.844026E-16 | 1.999999E+00  | 2.000001E+00 | 1.725095E-06 | 1.618034                              |
| 36  | 2.000000E+00  | 2.000000E+00 | 3.844026E-16 | 5.382674E-14 | 2.000000E+00  | 2.000001E+00 | 1.066366E-06 | 1.617732                              |
| 37  | 2.000000E+00  | 2.000000E+00 | 3.073351E-14 | 3.844026E-16 | 2.000000E+00  | 2.000000E+00 | 6.590506E-07 | 1.618034                              |
| 38  | 2.000000E+00  | 2.000000E+00 | 3.844026E-16 | 5.840808E-15 | 2.000000E+00  | 2.000000E+00 | 4.073157E-07 | 1.618034                              |
| 39  | 2.000000E+00  | 2.000000E+00 | 6.265603E-15 | 3.844026E-16 | 2.000000E+00  | 2.000000E+00 | 2.517349E-07 | 1.618034                              |
| 40  | 2.000000E+00  | 2.000000E+00 | 3.844026E-16 | 2.889532E-16 | 2.000000E+00  | 2.000000E+00 | 1.555807E-07 | 1.618034                              |

### Метод Фибоначчи для $y(x) = (x - 9)^2$ :

| $i$ | $x_1$        | $x_2$        | $f(x_1)$     | $f(x_2)$     | $a_i$         | $b_i$        | $b_i - a_i$  | $\frac{b_{i-1} - a_{i-1}}{b_i - a_i}$ |
|-----|--------------|--------------|--------------|--------------|---------------|--------------|--------------|---------------------------------------|
| 1   | 6.403252E+00 | 1.159675E+01 | 6.743099E+00 | 6.743099E+00 | -2.000000E+00 | 2.000000E+01 | 2.200000E+01 | 0                                     |
| 2   | 1.159675E+01 | 1.480650E+01 | 6.743099E+00 | 3.371549E+01 | 6.403252E+00  | 2.000000E+01 | 1.359675E+01 | 1.618034                              |
| 3   | 9.613009E+00 | 1.159675E+01 | 3.757800E-01 | 6.743099E+00 | 6.403252E+00  | 1.480650E+01 | 8.403252E+00 | 1.618034                              |
| 4   | 8.386991E+00 | 9.613009E+00 | 3.757800E-01 | 3.757800E-01 | 6.403252E+00  | 1.159675E+01 | 5.193496E+00 | 1.618034                              |
| 5   | 9.613009E+00 | 1.037073E+01 | 3.757800E-01 | 1.878900E+00 | 8.386991E+00  | 1.159675E+01 | 3.209757E+00 | 1.618034                              |
| 6   | 9.144712E+00 | 9.613009E+00 | 2.094150E-02 | 3.757800E-01 | 8.386991E+00  | 1.037073E+01 | 1.983739E+00 | 1.618034                              |
| 7   | 8.855288E+00 | 9.144712E+00 | 2.094150E-02 | 2.094150E-02 | 8.386991E+00  | 9.613009E+00 | 1.226018E+00 | 1.618034                              |
| 8   | 9.144712E+00 | 9.323585E+00 | 2.094150E-02 | 1.047075E-01 | 8.855288E+00  | 9.613009E+00 | 7.577208E-01 | 1.618034                              |
| 9   | 9.034162E+00 | 9.144712E+00 | 1.167030E-03 | 2.094150E-02 | 8.855288E+00  | 9.323585E+00 | 4.682972E-01 | 1.618034                              |
| 10  | 8.965838E+00 | 9.034162E+00 | 1.167030E-03 | 1.167030E-03 | 8.855288E+00  | 9.144712E+00 | 2.894236E-01 | 1.618034                              |
| 11  | 9.034162E+00 | 9.076388E+00 | 1.167030E-03 | 5.835150E-03 | 8.965838E+00  | 9.144712E+00 | 1.788736E-01 | 1.618034                              |
| 12  | 9.008065E+00 | 9.034162E+00 | 6.503635E-05 | 1.167030E-03 | 8.965838E+00  | 9.076388E+00 | 1.105500E-01 | 1.618034                              |
| 13  | 8.991935E+00 | 9.008065E+00 | 6.503635E-05 | 6.503635E-05 | 8.965838E+00  | 9.034162E+00 | 6.832364E-02 | 1.618034                              |
| 14  | 9.008065E+00 | 9.018033E+00 | 6.503635E-05 | 3.251818E-04 | 8.991935E+00  | 9.034162E+00 | 4.222633E-02 | 1.618034                              |
| 15  | 9.001904E+00 | 9.008065E+00 | 3.624352E-06 | 6.503635E-05 | 8.991935E+00  | 9.018033E+00 | 2.609731E-02 | 1.618034                              |
| 16  | 8.998096E+00 | 9.001904E+00 | 3.624352E-06 | 3.624352E-06 | 8.991935E+00  | 9.008065E+00 | 1.612902E-02 | 1.618034                              |
| 17  | 9.001904E+00 | 9.004257E+00 | 3.624352E-06 | 1.812176E-05 | 8.998096E+00  | 9.008065E+00 | 9.968285E-03 | 1.618034                              |
| 18  | 9.000449E+00 | 9.001904E+00 | 2.019782E-07 | 3.624352E-06 | 8.998096E+00  | 9.004257E+00 | 6.160739E-03 | 1.618034                              |
| 19  | 8.999551E+00 | 9.000449E+00 | 2.019782E-07 | 2.019782E-07 | 8.998096E+00  | 9.001904E+00 | 3.807546E-03 | 1.618034                              |
| 20  | 9.000449E+00 | 9.001005E+00 | 2.019782E-07 | 1.009891E-06 | 8.999551E+00  | 9.001904E+00 | 2.353193E-03 | 1.618034                              |
| 21  | 9.000106E+00 | 9.000449E+00 | 1.125586E-08 | 2.019782E-07 | 8.999551E+00  | 9.001005E+00 | 1.454353E-03 | 1.618034                              |
| 22  | 8.999894E+00 | 9.000106E+00 | 1.125586E-08 | 1.125586E-08 | 8.999551E+00  | 9.000449E+00 | 8.988397E-04 | 1.618034                              |
| 23  | 9.000106E+00 | 9.000237E+00 | 1.125586E-08 | 5.627929E-08 | 8.999894E+00  | 9.000449E+00 | 5.555135E-04 | 1.618034                              |
| 24  | 9.000025E+00 | 9.000106E+00 | 6.272673E-10 | 1.125586E-08 | 8.999894E+00  | 9.000237E+00 | 3.433262E-04 | 1.618034                              |
| 25  | 8.999975E+00 | 9.000025E+00 | 6.272673E-10 | 6.272673E-10 | 8.999894E+00  | 9.000106E+00 | 2.121873E-04 | 1.618034                              |
| 26  | 9.000025E+00 | 9.000056E+00 | 6.272673E-10 | 3.136339E-09 | 8.999975E+00  | 9.000106E+00 | 1.311389E-04 | 1.618034                              |
| 27  | 9.000006E+00 | 9.000025E+00 | 3.495668E-11 | 6.272673E-10 | 8.999975E+00  | 9.000056E+00 | 8.104833E-05 | 1.618034                              |
| 28  | 8.999994E+00 | 9.000006E+00 | 3.495668E-11 | 3.495668E-11 | 8.999975E+00  | 9.000025E+00 | 5.009061E-05 | 1.618034                              |
| 29  | 9.000006E+00 | 9.000013E+00 | 3.495668E-11 | 1.747808E-10 | 8.999994E+00  | 9.000025E+00 | 3.095772E-05 | 1.618033                              |
| 30  | 9.000001E+00 | 9.000006E+00 | 1.947797E-12 | 3.495668E-11 | 8.999994E+00  | 9.000013E+00 | 1.913289E-05 | 1.618037                              |
| 31  | 8.999999E+00 | 9.000001E+00 | 1.947797E-12 | 1.947797E-12 | 8.999994E+00  | 9.000006E+00 | 1.182483E-05 | 1.618026                              |
| 32  | 9.000001E+00 | 9.000003E+00 | 1.947797E-12 | 9.741561E-12 | 8.999999E+00  | 9.000006E+00 | 7.308052E-06 | 1.618056                              |
| 33  | 9.000000E+00 | 9.000001E+00 | 1.088191E-13 | 1.947797E-12 | 8.999999E+00  | 9.000003E+00 | 4.516782E-06 | 1.617978                              |
| 34  | 9.000000E+00 | 9.000000E+00 | 1.088191E-13 | 1.088191E-13 | 8.999999E+00  | 9.000001E+00 | 2.791270E-06 | 1.618182                              |
| 35  | 9.000000E+00 | 9.000001E+00 | 1.088191E-13 | 5.415198E-13 | 9.000000E+00  | 9.000001E+00 | 1.725512E-06 | 1.617647                              |
| 36  | 9.000000E+00 | 9.000000E+00 | 5.795098E-15 | 1.088191E-13 | 9.000000E+00  | 9.000001E+00 | 1.065758E-06 | 1.619048                              |
| 37  | 9.000000E+00 | 9.000000E+00 | 5.795098E-15 | 5.795098E-15 | 9.000000E+00  | 9.000000E+00 | 6.597547E-07 | 1.615385                              |
| 38  | 9.000000E+00 | 9.000000E+00 | 5.795098E-15 | 3.155109E-14 | 9.000000E+00  | 9.000000E+00 | 4.060029E-07 | 1.625                                 |
| 39  | 9.000000E+00 | 9.000000E+00 | 6.438997E-16 | 5.795098E-15 | 9.000000E+00  | 9.000000E+00 | 2.537518E-07 | 1.6                                   |
| 40  | 9.000000E+00 | 9.000000E+00 | 6.439000E-16 | 6.438997E-16 | 9.000000E+00  | 9.000000E+00 | 1.522511E-07 | 1.666667                              |
| 41  | 9.000000E+00 | 9.000000E+00 | 6.438997E-16 | 6.438997E-16 | 9.000000E+00  | 9.000000E+00 | 1.015007E-07 | 1.5                                   |

### Метод Фибоначчи для $y(x) = (x - 2)^2$ :

| $i$ | $x_1$         | $x_2$        | $f(x_1)$     | $f(x_2)$     | $a_i$         | $b_i$        | $b_i - a_i$  | $\frac{b_{i-1} - a_{i-1}}{b_i - a_i}$ |
|-----|---------------|--------------|--------------|--------------|---------------|--------------|--------------|---------------------------------------|
| 1   | 6.403252E+00  | 1.159675E+01 | 1.938863E+01 | 9.209757E+01 | -2.000000E+00 | 2.000000E+01 | 2.200000E+01 | 0                                     |
| 2   | 3.193496E+00  | 6.403252E+00 | 1.424432E+00 | 1.938863E+01 | -2.000000E+00 | 1.159675E+01 | 1.359675E+01 | 1.618034                              |
| 3   | 1.209757E+00  | 3.193496E+00 | 6.244844E-01 | 1.424432E+00 | -2.000000E+00 | 6.403252E+00 | 8.403252E+00 | 1.618034                              |
| 4   | -1.626124E-02 | 1.209757E+00 | 4.065309E+00 | 6.244844E-01 | -2.000000E+00 | 3.193496E+00 | 5.193496E+00 | 1.618034                              |
| 5   | 1.209757E+00  | 1.967478E+00 | 6.244844E-01 | 1.057711E-03 | -1.626124E-02 | 3.193496E+00 | 3.209757E+00 | 1.618034                              |
| 6   | 1.967478E+00  | 2.435775E+00 | 1.057711E-03 | 1.898996E-01 | 1.209757E+00  | 3.193496E+00 | 1.983739E+00 | 1.618034                              |
| 7   | 1.678054E+00  | 1.967478E+00 | 1.036493E-01 | 1.057711E-03 | 1.209757E+00  | 2.435775E+00 | 1.226018E+00 | 1.618034                              |
| 8   | 1.967478E+00  | 2.146351E+00 | 1.057711E-03 | 2.141866E-02 | 1.678054E+00  | 2.435775E+00 | 7.577208E-01 | 1.618034                              |
| 9   | 1.856928E+00  | 1.967478E+00 | 2.046973E-02 | 1.057711E-03 | 1.678054E+00  | 2.146351E+00 | 4.682972E-01 | 1.618034                              |
| 10  | 1.967478E+00  | 2.035801E+00 | 1.057711E-03 | 1.281723E-03 | 1.856928E+00  | 2.146351E+00 | 2.894236E-01 | 1.618034                              |
| 11  | 1.925251E+00  | 1.967478E+00 | 5.587384E-03 | 1.057711E-03 | 1.856928E+00  | 2.035801E+00 | 1.788736E-01 | 1.618034                              |
| 12  | 1.967478E+00  | 1.993575E+00 | 1.057711E-03 | 4.128277E-05 | 1.925251E+00  | 2.035801E+00 | 1.105500E-01 | 1.618034                              |
| 13  | 1.993575E+00  | 2.009704E+00 | 4.128277E-05 | 9.416484E-05 | 1.967478E+00  | 2.035801E+00 | 6.832364E-02 | 1.618034                              |
| 14  | 1.983607E+00  | 1.993575E+00 | 2.687452E-04 | 4.128277E-05 | 1.967478E+00  | 2.009704E+00 | 4.222633E-02 | 1.618034                              |
| 15  | 1.993575E+00  | 1.999736E+00 | 4.128277E-05 | 6.992209E-08 | 1.983607E+00  | 2.009704E+00 | 2.609731E-02 | 1.618034                              |
| 16  | 1.999736E+00  | 2.003543E+00 | 6.992209E-08 | 1.255369E-05 | 1.993575E+00  | 2.009704E+00 | 1.612902E-02 | 1.618034                              |
| 17  | 1.997382E+00  | 1.999736E+00 | 6.851938E-06 | 6.992209E-08 | 1.993575E+00  | 2.003543E+00 | 9.968285E-03 | 1.618034                              |
| 18  | 1.999736E+00  | 2.001190E+00 | 6.992209E-08 | 1.415922E-06 | 1.997382E+00  | 2.003543E+00 | 6.160739E-03 | 1.618034                              |
| 19  | 1.998837E+00  | 1.999736E+00 | 1.353191E-06 | 6.992209E-08 | 1.997382E+00  | 2.001190E+00 | 3.807546E-03 | 1.618034                              |
| 20  | 1.999736E+00  | 2.000291E+00 | 6.992209E-08 | 8.473084E-08 | 1.998837E+00  | 2.001190E+00 | 2.353193E-03 | 1.618034                              |
| 21  | 1.999392E+00  | 1.999736E+00 | 3.693650E-07 | 6.992209E-08 | 1.998837E+00  | 2.000291E+00 | 1.454353E-03 | 1.618034                              |
| 22  | 1.999736E+00  | 1.999948E+00 | 6.992209E-08 | 2.729078E-09 | 1.999392E+00  | 2.000291E+00 | 8.988397E-04 | 1.618034                              |
| 23  | 1.999948E+00  | 2.000079E+00 | 2.729078E-09 | 6.224951E-09 | 1.999736E+00  | 2.000291E+00 | 5.555135E-04 | 1.618034                              |
| 24  | 1.999867E+00  | 1.999948E+00 | 1.776593E-08 | 2.729078E-09 | 1.999736E+00  | 2.000079E+00 | 3.433262E-04 | 1.618034                              |
| 25  | 1.999948E+00  | 1.999998E+00 | 2.729078E-09 | 4.622371E-12 | 1.999867E+00  | 2.000079E+00 | 2.121873E-04 | 1.618034                              |
| 26  | 1.999998E+00  | 2.000029E+00 | 4.622371E-12 | 8.298865E-10 | 1.999948E+00  | 2.000079E+00 | 1.311389E-04 | 1.618034                              |
| 27  | 1.999979E+00  | 1.999998E+00 | 4.529600E-10 | 4.622371E-12 | 1.999948E+00  | 2.000029E+00 | 8.104833E-05 | 1.618034                              |
| 28  | 1.999998E+00  | 2.000010E+00 | 4.622371E-12 | 9.360301E-11 | 1.999979E+00  | 2.000029E+00 | 5.009061E-05 | 1.618034                              |
| 29  | 1.999991E+00  | 1.999998E+00 | 8.945418E-11 | 4.622371E-12 | 1.999979E+00  | 2.000010E+00 | 3.095772E-05 | 1.618033                              |
| 30  | 1.999998E+00  | 2.000002E+00 | 4.622371E-12 | 5.601801E-12 | 1.999991E+00  | 2.000010E+00 | 1.913289E-05 | 1.618037                              |
| 31  | 1.999995E+00  | 1.999998E+00 | 2.441585E-11 | 4.622371E-12 | 1.999991E+00  | 2.000002E+00 | 1.182483E-05 | 1.618026                              |
| 32  | 1.999998E+00  | 2.000000E+00 | 4.622371E-12 | 1.801642E-13 | 1.999995E+00  | 2.000002E+00 | 7.308052E-06 | 1.618056                              |

|    |              |              |              |              |              |              |              |          |
|----|--------------|--------------|--------------|--------------|--------------|--------------|--------------|----------|
| 33 | 2.000000E+00 | 2.000001E+00 | 1.801642E-13 | 4.112657E-13 | 1.999998E+00 | 2.000002E+00 | 4.516782E-06 | 1.617978 |
| 34 | 1.999999E+00 | 2.000000E+00 | 1.175516E-12 | 1.801642E-13 | 1.999998E+00 | 2.000001E+00 | 2.791270E-06 | 1.618182 |
| 35 | 2.000000E+00 | 2.000000E+00 | 1.801642E-13 | 3.405751E-16 | 1.999999E+00 | 2.000001E+00 | 1.725512E-06 | 1.617647 |
| 36 | 2.000000E+00 | 2.000000E+00 | 3.405751E-16 | 5.536474E-14 | 2.000000E+00 | 2.000001E+00 | 1.065758E-06 | 1.619048 |
| 37 | 2.000000E+00 | 2.000000E+00 | 2.914046E-14 | 3.405751E-16 | 2.000000E+00 | 2.000000E+00 | 6.597547E-07 | 1.615385 |
| 38 | 2.000000E+00 | 2.000000E+00 | 3.405751E-16 | 6.896646E-15 | 2.000000E+00 | 2.000000E+00 | 4.060029E-07 | 1.625000 |
| 39 | 2.000000E+00 | 2.000000E+00 | 4.789338E-15 | 3.405751E-16 | 2.000000E+00 | 2.000000E+00 | 2.537518E-07 | 1.600000 |
| 40 | 2.000000E+00 | 2.000000E+00 | 3.405751E-16 | 1.043011E-15 | 2.000000E+00 | 2.000000E+00 | 1.522511E-07 | 1.666667 |
| 41 | 2.000000E+00 | 2.000000E+00 | 3.405751E-16 | 3.405751E-16 | 2.000000E+00 | 2.000000E+00 | 1.015007E-07 | 1.500000 |

График полученной зависимости для  $y(x) = (x - 9)^2$ :

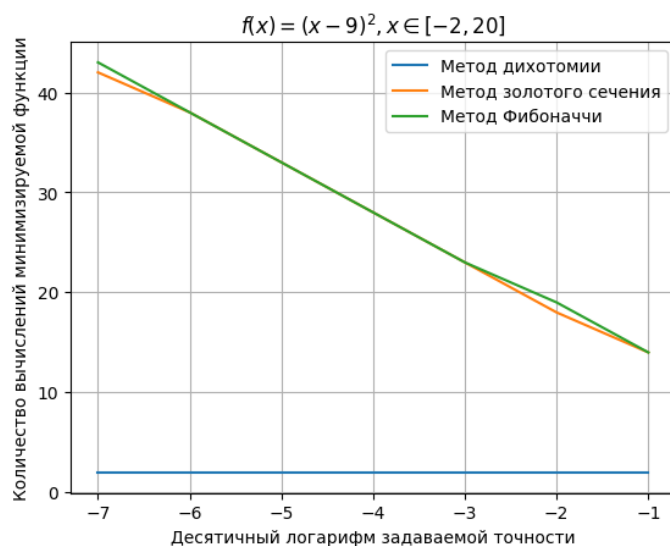
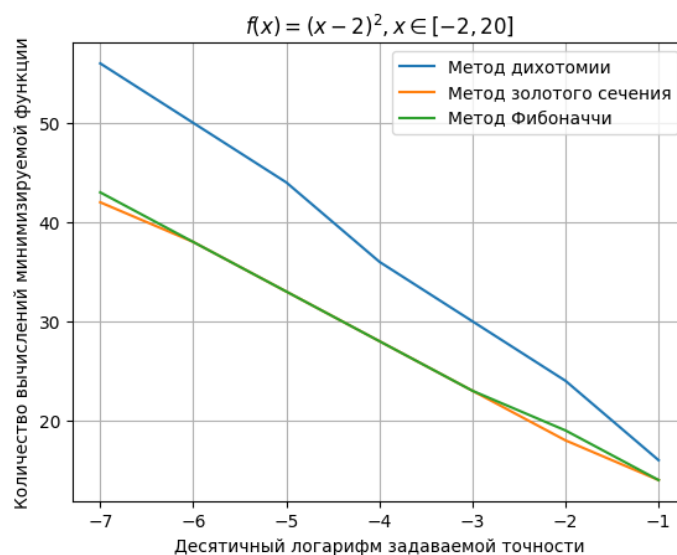


График полученной зависимости для  $y(x) = (x - 2)^2$ :



Поиск интервала, содержащего минимум функции:

| i                                                          | $x_i$        | $f(x_i)$     |
|------------------------------------------------------------|--------------|--------------|
| 1                                                          | 3.000000E-10 | 8.100000E+01 |
| 2                                                          | 7.000000E-10 | 8.100000E+01 |
| 3                                                          | 1.500000E-09 | 8.100000E+01 |
| 4                                                          | 3.100000E-09 | 8.100000E+01 |
| 5                                                          | 6.300000E-09 | 8.100000E+01 |
| 6                                                          | 1.270000E-08 | 8.100000E+01 |
| 7                                                          | 2.550000E-08 | 8.100000E+01 |
| 8                                                          | 5.110000E-08 | 8.100000E+01 |
| 9                                                          | 1.023000E-07 | 8.100000E+01 |
| 10                                                         | 2.047000E-07 | 8.100000E+01 |
| 11                                                         | 4.095000E-07 | 8.099999E+01 |
| 12                                                         | 8.191000E-07 | 8.099999E+01 |
| 13                                                         | 1.638300E-06 | 8.099997E+01 |
| 14                                                         | 3.276700E-06 | 8.099994E+01 |
| 15                                                         | 6.553500E-06 | 8.099988E+01 |
| 16                                                         | 1.310710E-05 | 8.099976E+01 |
| 17                                                         | 2.621430E-05 | 8.099953E+01 |
| 18                                                         | 5.242870E-05 | 8.099906E+01 |
| 19                                                         | 1.048575E-04 | 8.099811E+01 |
| 20                                                         | 2.097151E-04 | 8.099623E+01 |
| 21                                                         | 4.194303E-04 | 8.099245E+01 |
| 22                                                         | 8.388607E-04 | 8.098490E+01 |
| 23                                                         | 1.677722E-03 | 8.096980E+01 |
| 24                                                         | 3.355443E-03 | 8.093961E+01 |
| 25                                                         | 6.710886E-03 | 8.087925E+01 |
| 26                                                         | 1.342177E-02 | 8.075859E+01 |
| 27                                                         | 2.684355E-02 | 8.051754E+01 |
| 28                                                         | 5.368709E-02 | 8.003651E+01 |
| 29                                                         | 1.073742E-01 | 7.907879E+01 |
| 30                                                         | 2.147484E-01 | 7.718065E+01 |
| 31                                                         | 4.294967E-01 | 7.345353E+01 |
| 32                                                         | 8.589935E-01 | 6.627599E+01 |
| 33                                                         | 1.717987E+00 | 5.302771E+01 |
| 34                                                         | 3.435974E+00 | 3.095839E+01 |
| 35                                                         | 6.871948E+00 | 4.528607E+00 |
| 36                                                         | 1.374390E+01 | 2.250454E+01 |
| Интервал, содержащий минимум: [3.435974E+00, 1.374390E+01] |              |              |

## Выводы

Метод дихотомии в нашем случае нашёл минимум функции за одну итерацию по случайности, поэтому полученный результат не отражает эффективности метода.

Эффективность метода золотого сечения совпала с теоретической:

$$40 = n \geq \frac{\ln\left(\frac{b-a}{\varepsilon}\right)}{\ln\left(\frac{\sqrt{5}+1}{2}\right)} = \frac{\ln\left(\frac{22+2}{1e-7}\right)}{\ln(1.61803)} = \frac{19.29615}{0.48121} = 40.09923$$

Эффективность метода Фибоначчи близка к эффективности метода золотого сечения.